Can we see a missed payment coming?

Predicting credit card default · AAI-500 Final Team Project

Team 10 — Prasad Pidaparty · Siddhant Bhura · Eshwar Reddy M · Umesh Sharma

The decision we are supporting

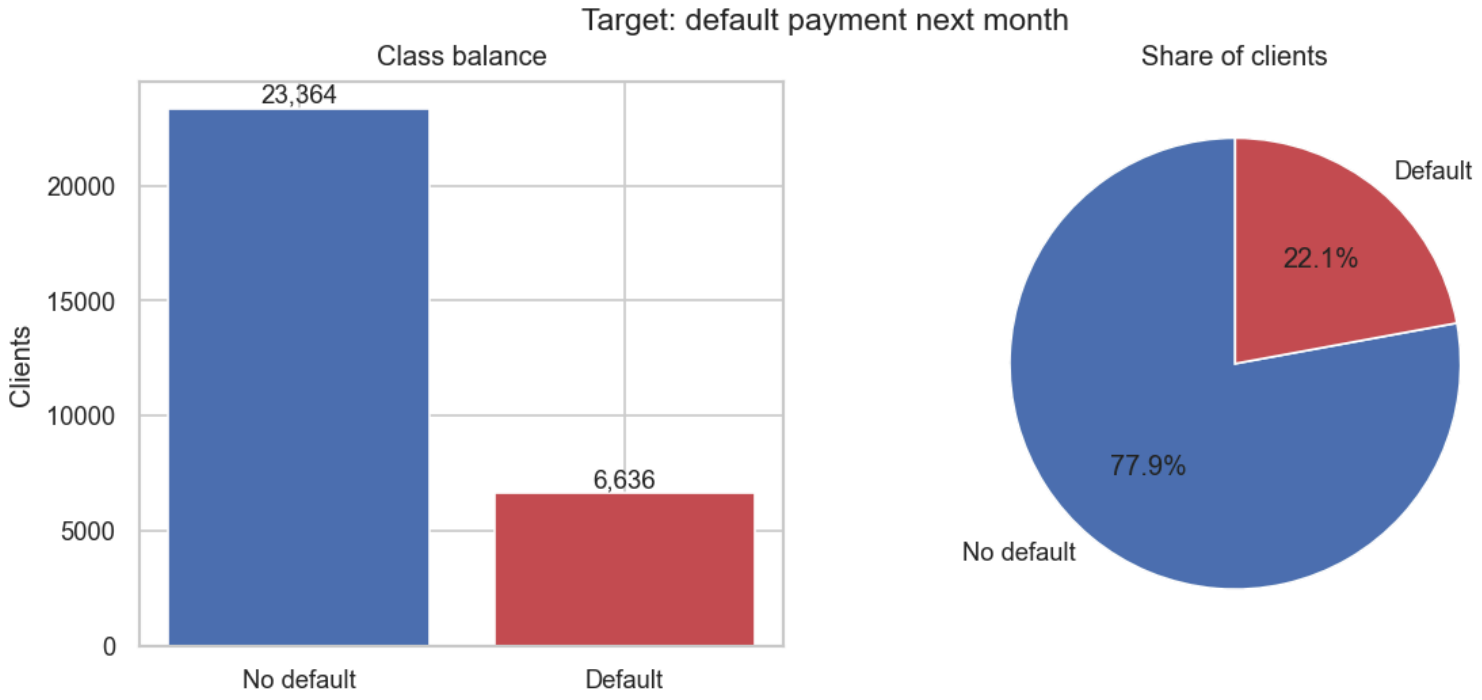
Not a verdict on a customer — a ranked list for a team with limited time

- A collections team can call a fixed number of customers each month
- Calling the wrong ones wastes the budget; missing the right ones costs the balance
- So the model's job is to ORDER customers by risk, not to pass judgement
- Every design choice in this project follows from that one sentence

What we looked at

30,000 customers · six months of history · one in five missed the next payment

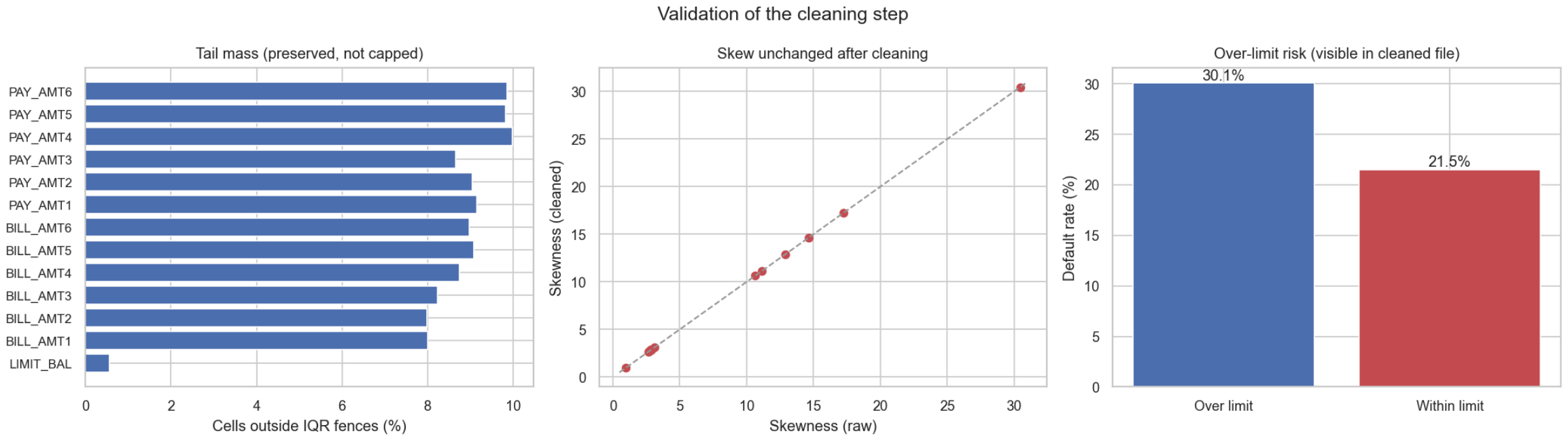
- Public UCI dataset from a Taiwanese bank, April to September 2005
- Credit limit, demographics, six monthly bills, six monthly payments, six lateness codes
- 6,636 of 30,000 customers defaulted — 22.1%
- That imbalance sets a trap we will come back to at the end



Is the data trustworthy?

We checked rather than assumed — and we checked our own cleaning too

- No missing values and no duplicate records anywhere in 30,000 rows
- Fixed: headings on the wrong row, a misnamed column, undocumented category codes
- Extreme amounts were kept, not capped, so over-limit customers stay visible
- Re-ran every conclusion on the untouched raw file — the answers did not move

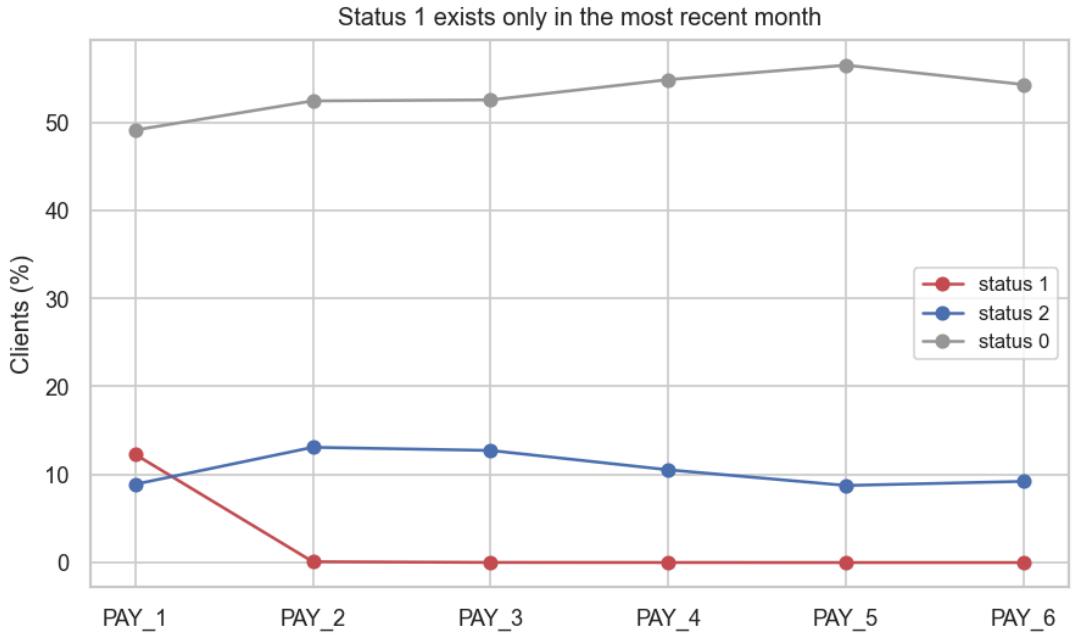


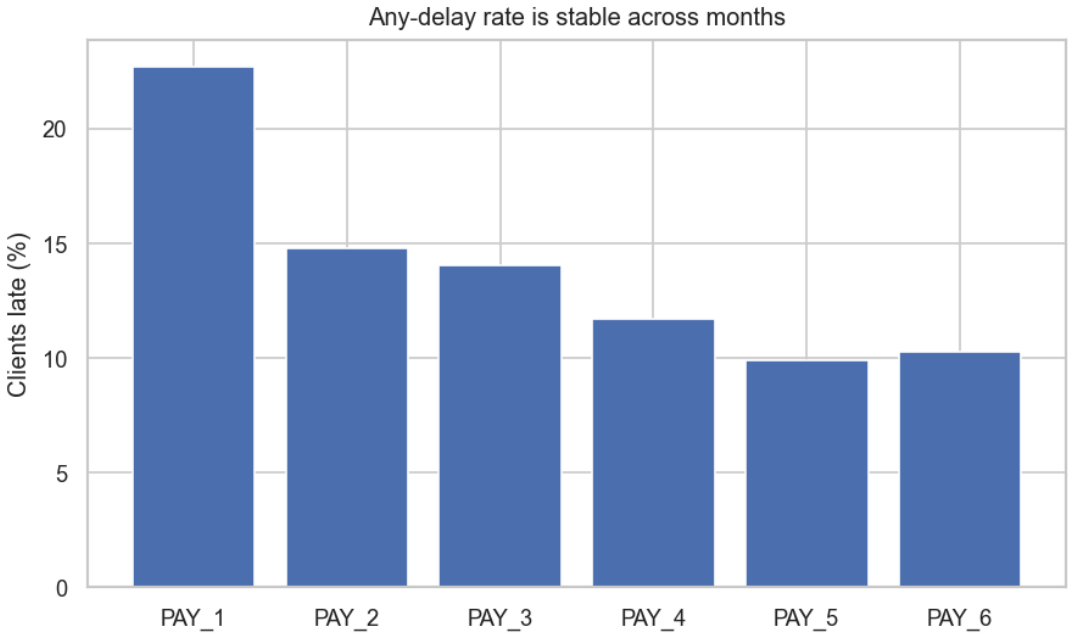
The flaw we found before the model did

The six monthly lateness codes are not the same ruler

- "One month late" appears 3,688 times in September and ZERO times in April and May
- A naive trend feature says customers who "improved" default at 26% vs 16% for stable
- They never improved — the code changed meaning underneath them
- So we built no trend feature, and used only measures immune to the problem

The repayment scale shifts between months



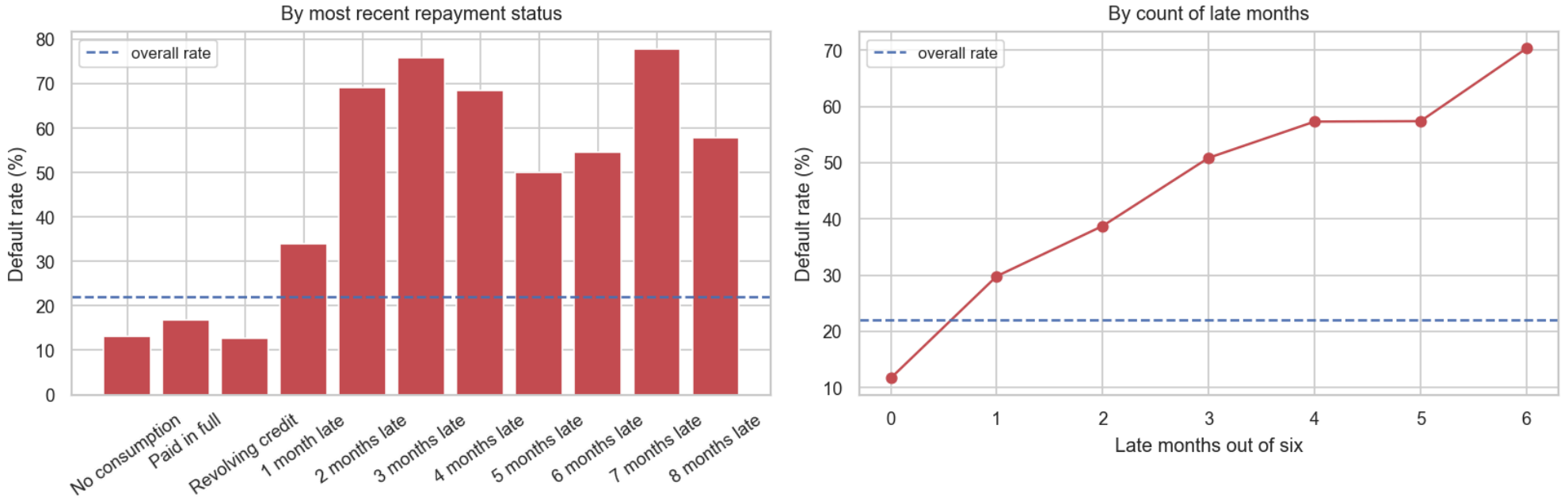


What actually predicts a missed payment

Recent payment behaviour, and almost nothing else

- Current on the last payment: 13.8% default. One month late: 34%. Two months: 69%
- Count of late months runs 11.7% (never) to 70.3% (all six) with no reversals
- Paid nothing against the bill: 51.8%. Cleared it: 13.9%
- Late recently 41.6% vs late long ago 26.0% — recency carries ~1.6x the risk

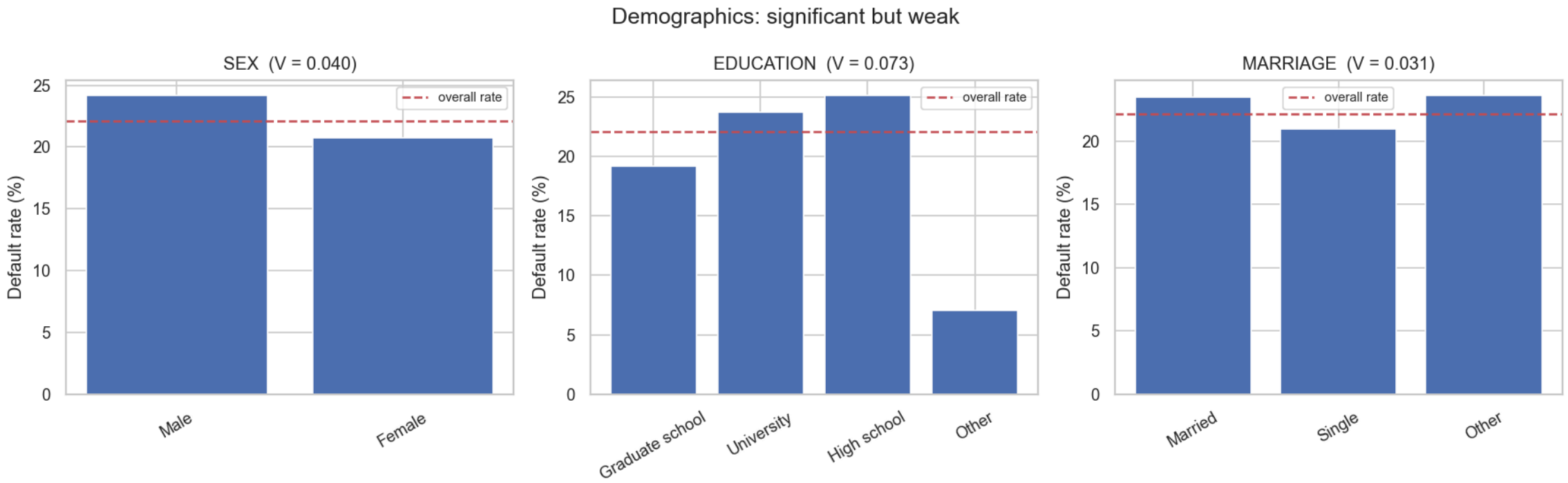
Repayment history dominates every other signal



What does NOT predict it

Statistically significant, practically useless — and a fairness risk

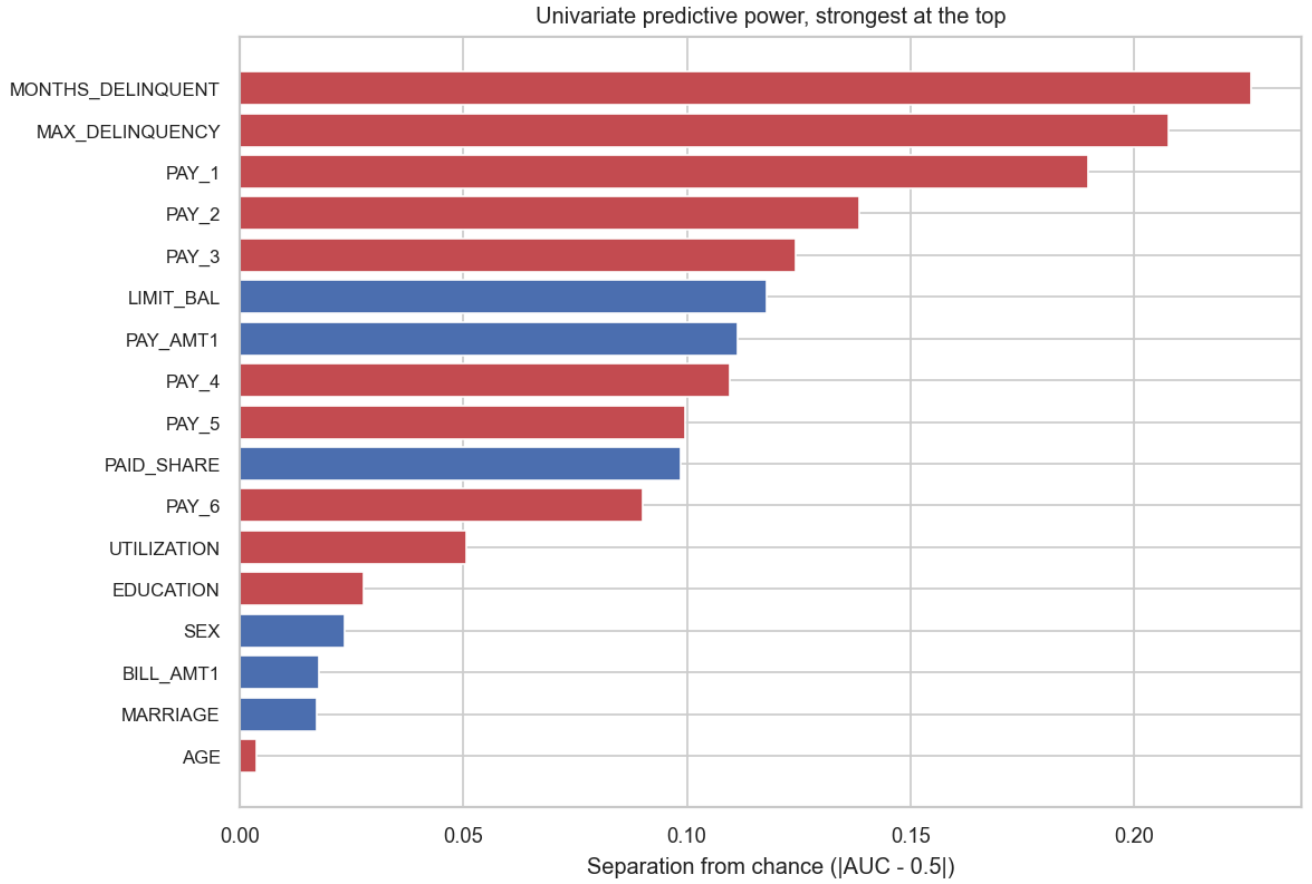
- Age, sex, education, marital status: all significant at $p < .001$, all effect sizes under 0.08
- Age on its own scores 0.504 — indistinguishable from a coin flip
- Because risk is U-shaped: 26.7% in the early 20s, 19.4% in the low 30s, 26.8% past 60
- We excluded demographics from the final model entirely



Ranking every signal honestly

The dataset does not ship with its own best predictor — we had to build it

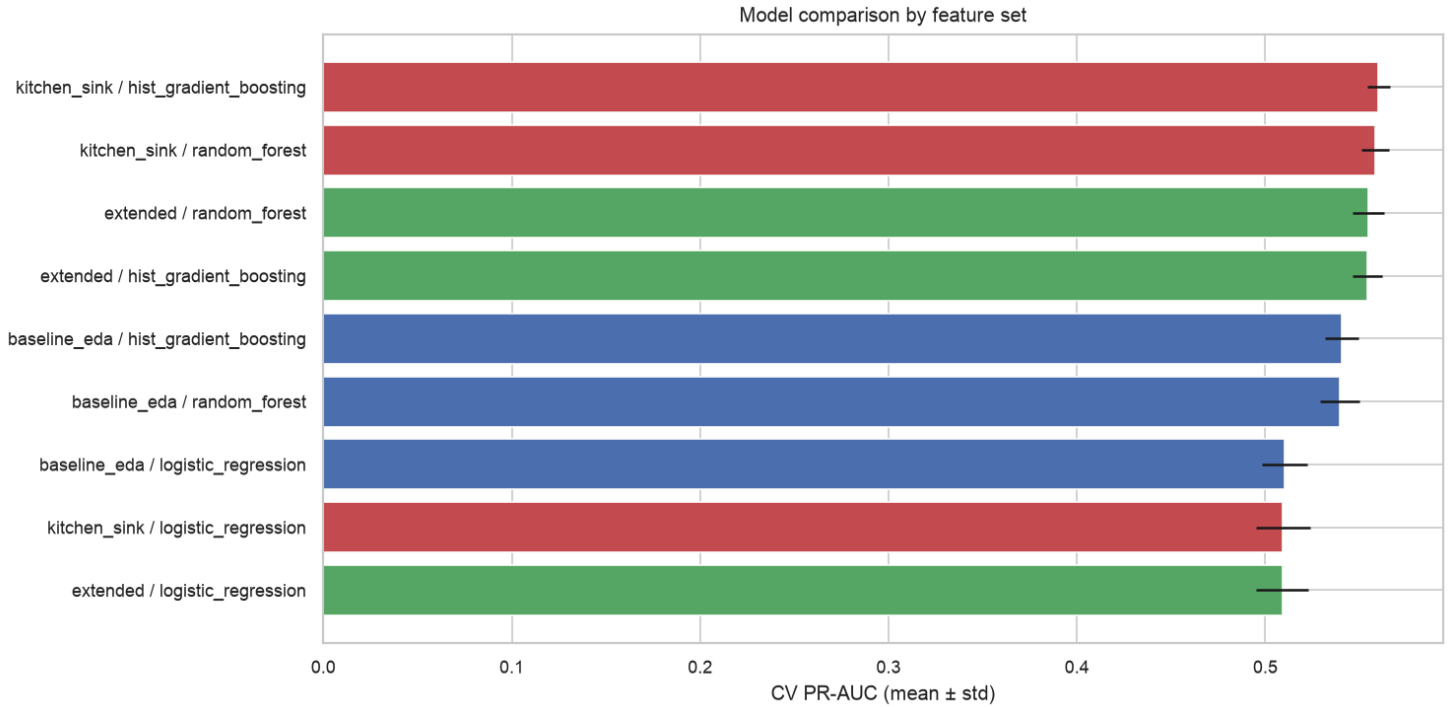
- Each variable scored alone, on its own, against the outcome
- Best single signal is the COUNT of late months, at 0.726 — and it is engineered, not given
- The repayment block fills the top of the list; demographics fill the bottom
- The six bill columns repeat each other (correlations 0.80–0.95), so we keep one



How we chose the model

Nine combinations, judged on ranking quality — never on accuracy

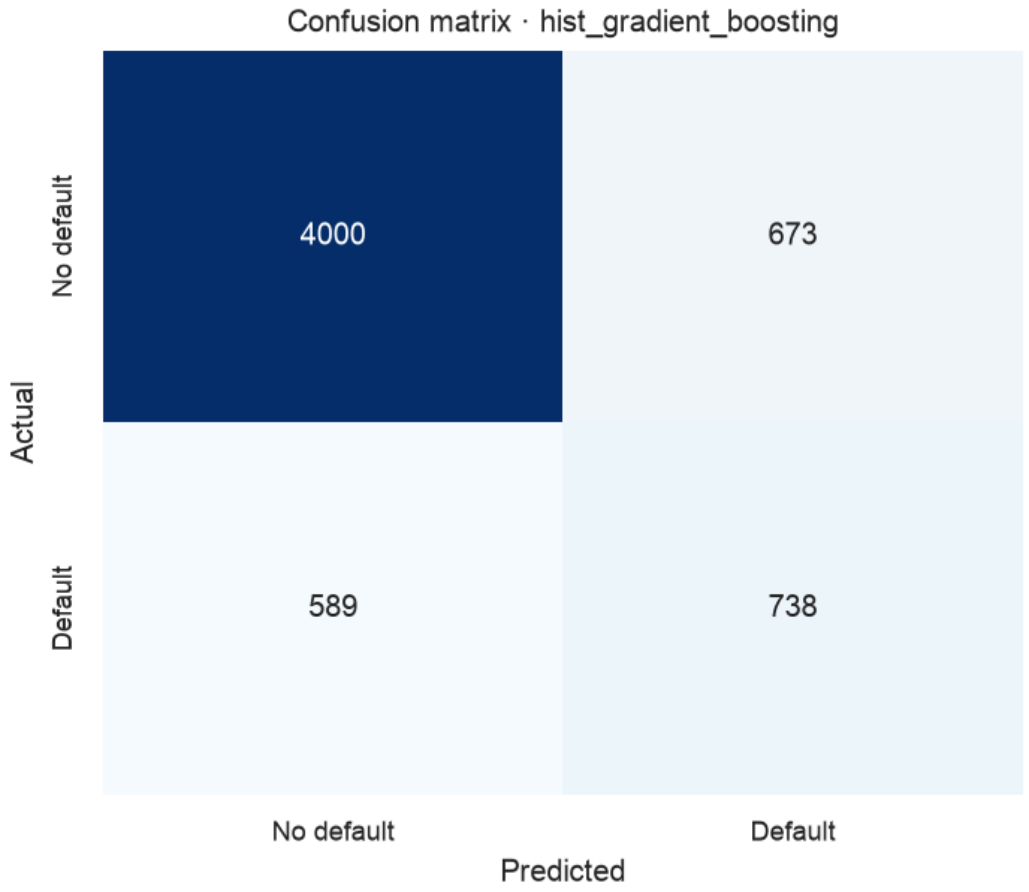
- Three model families × three input sets, each validated on data it had never seen
- A transparent statistical model, two tree-based methods, plus a neural network
- Scored on how well each ORDERS customers by risk, not on accuracy
- Tree-based methods won; the top few are close enough to be a practical tie



How well it works

On 6,000 customers the model had never seen

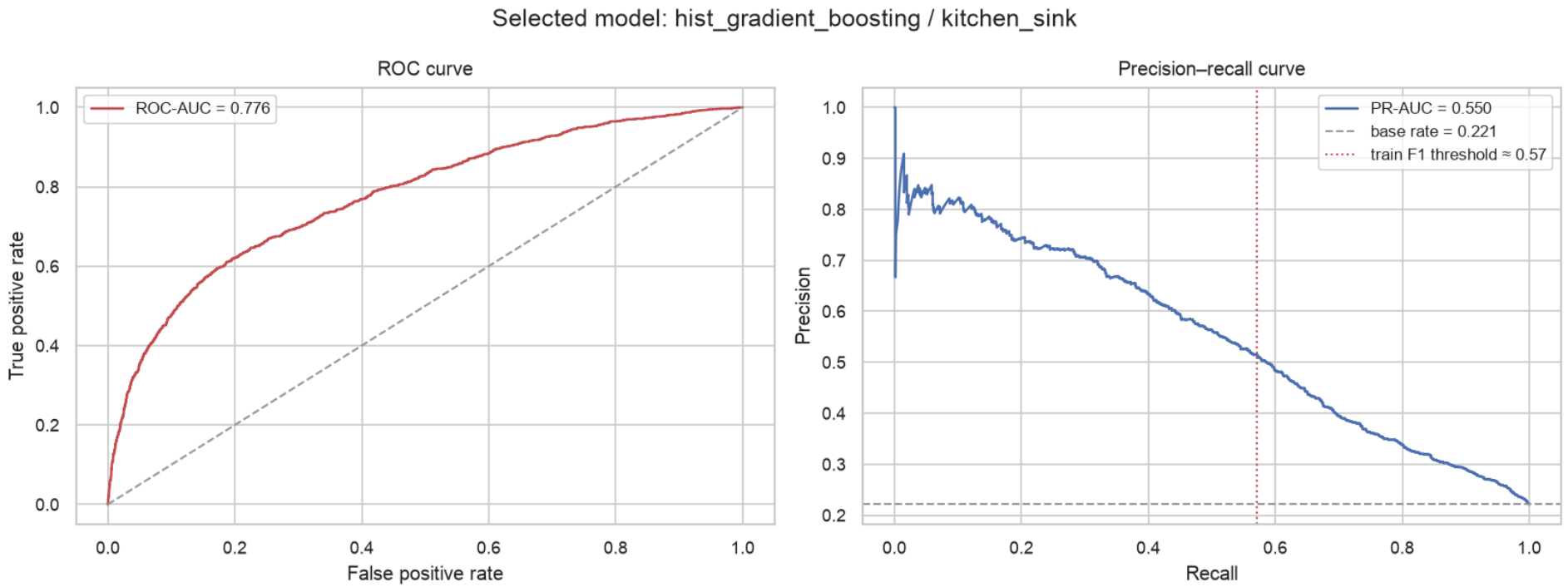
- Found 738 of 1,327 real defaulters — 55.6%
- Flagged 1,411 customers; 738 of them genuinely defaulted — 52.3%
- Flagging 1,411 AT RANDOM would have caught about 312
- So the same amount of effort catches 2.4x as many real defaulters



The trap, and what the model cannot do

Why we never once reported accuracy

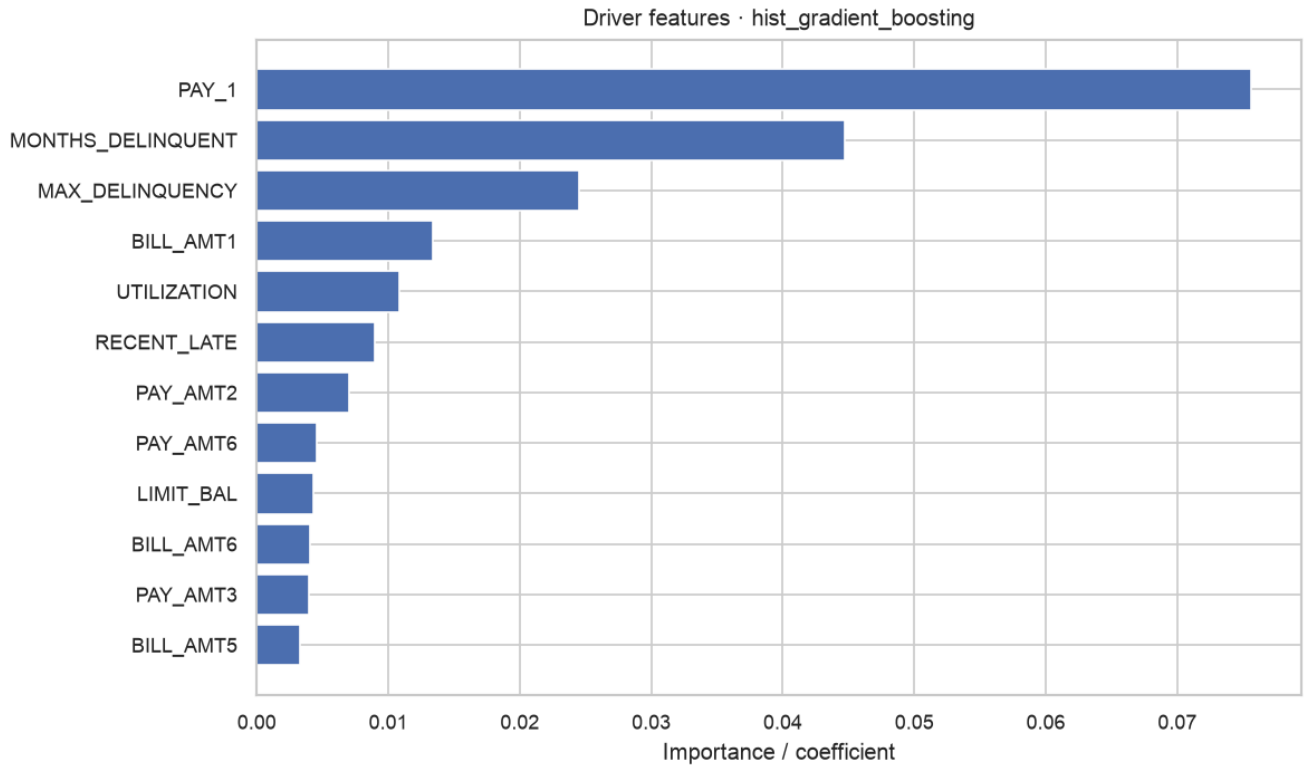
- Our accuracy is 79.0%. Always answering "no default" scores 77.9%
- By the measure most people reach for first, the model is barely worth running
- By the measure that matches the decision, it is clearly worth running
- It cannot say a given customer WILL default — 70% flagged still has a 30% chance of paying



What we recommend

Four things a lender should do with this

- Deploy it as a ranked monthly outreach list — never as an automated decision
- Re-set the cut-off from the bank's own costs; ours is tuned for statistical balance
- Fix the lateness coding — it would make five more months of history usable
- Treat "two months behind" as a trigger on its own; that group defaults more often than not



Who did what

Team 10 · contributions to the final deliverables

Team member	Contribution
Siddhant Bhura	Data cleaning and preparation · neural network analysis · technical report
Eshwar Reddy M	Exploratory data analysis · regression assumption testing · presentation
Umesh Sharma	Model development, comparison and validation · reproducibility pipeline
Prasad Pidaparty	Dataset selection and business problem definition (project plan stage)

Thank you — questions welcome